Jack Pham

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EDUCATION

University of California, Berkeley — **Berkeley, CA**
Bachelor of Science in Electrical Engineering and Computer Sciences, GPA: 3.54 — *August 2025 - May 2028 (Expected)*

Cypress College — **Cypress, CA**
Associate Degree for Transfer in Mathematics and Physics, GPA: 3.94 — *February 2023 - May 2025*

TECHNICAL SKILLS

Circuit & IC Design: RF and Analog Circuits, LTspice, PCB Design (KiCad), Power Regulation, Hardware Debug (oscilloscopes, multimeters, network and spectrum analyzers), Failure Analysis

Digital & Embedded: SystemVerilog, Verilog, cocotb, Icarus Verilog, Python Golden Models, Embedded C++ (ESP32-S3), I2S / I2C Sensor Interfacing

Programming & Tools: Python, C++, Rust, Git, Linux, Unix, Bash, JMP, Exensio, HPC Systems

WORK EXPERIENCE

Skyworks Solutions — **Irvine, CA**
Product Engineer Co-Op — *June 2026 - Present*

- Characterize RF chip performance on GaAs HBT and SAW power amplifier and low-noise amplifier devices using network and spectrum analyzers, measuring insertion loss, gain, and harmonic distortion during Final Test.
- Investigate product and manufacturing issues across semiconductor die, wafer, and module test data, applying Cpk, SPC, and ANOVA yield-analysis methods to isolate yield trends, process shifts, and root causes of performance failures on high-volume RF front-end products.
- Build and maintain the ARE limit-evaluation tool and STDF-to-CSV pipelines in Python, C++, and Rust, processing 3,000+ rows of production test data across seven-plus vendor formats while collaborating with 8 engineers across design, test, manufacturing, and quality.

The LEAP Institute (Huron Climate Resiliency Village / Su Mercadito) — **Huron, CA**
Volunteer Electrical Engineer, Solar + Storage Microgrid Design Support — *October 2026 - Present*

- Volunteer as electrical engineering support for a community-led solar, battery storage and EV charging microgrid serving a farmworker community in Huron, CA, working with graduate students at UCLA and USF and the LEAP Institute team.
- Review concept papers, budgets and a community survey (33 residents, 24 vendors) to identify electrical load drivers: refrigerated storage, a cooling center, a commissary kitchen, event lighting and stage power.
- Identify a conflict between the net zero goal and the planned DC fast charging (6 DCFC) and flag it to the team so they can define the net zero boundary.

RESEARCH EXPERIENCE

Texas Advanced Computing Center (TACC), UT Austin — **Austin, TX**
Summer Researcher — *June 2024 - August 2024*

- Formulated multi-robot task allocation as a linear program minimizing the maximum task-load discrepancy across 5 heterogeneous agents (Scout, Medic, Engineer, Carrier, FireBot) in a simulated disaster-response grid, combining it with a Fairness-Enhanced MADDPG reward function to improve task-distribution efficiency by 70% over the baseline.

PROJECTS

serdes-link: High-Speed Serial Link Model and RTL - Python, SystemVerilog, cocotb, KiCad — **Fall 2026**

- Reimplemented a PRBS generator, a 4-tap DFE and a CDR loop filter in SystemVerilog and verified them against Python golden models using cocotb and Icarus Verilog, with 40 automated tests (37 Python, 3 RTL cosimulations).
- Designed a multi-layer KiCad test-channel board to measure channel loss against the model; the board is designed but not yet fabricated.

AccessiSound: Sound Assistive Device | GitHub - Embedded C++, TensorFlow Lite Micro, ESP32-S3 — **Spring 2026**

- Engineered a $17 offline assistive device on ESP32-S3 that extracts MFCC features (Hamming windowing, Mel filterbank) to detect doorbells, fire alarms, and other household sounds for visually impaired users.

Methane Emission Monitoring Platform, Fung Fellowship | Fung Institute — **UC Berkeley, Spring 2026**

- Co-designed a $15 household methane sensor from off-the-shelf components and conducted 12 stakeholder interviews with Climate Action California to close a data gap in residential-level methane tracking.
- Worked with a 4-person cross-disciplinary team to fold stakeholder feedback into the sensor design and submitted findings to the California Air Resources Board to inform statewide methane mitigation policy.

UNIVERSITY ACTIVITIES

Underwater Robotics Team, UC Berkeley — **Berkeley, CA**
Electrical Subteam Member — *February 2026 - Present*

- Investigate power dissipation and thermal behavior of NVIDIA Jetson development modules used in autonomous underwater robots, and support electrical subsystem integration including power regulation, embedded hardware interfacing, and system wiring.
- Assist with debugging and testing of onboard computing systems connected to 16 sensors and actuators, and collaborate with the mechanical subteam to design and prepare 3D-printed fixtures and enclosures optimized for space, waterproofing, weight, and thermal performance.

HONORS & PRESENTATIONS

- PEARC26 Student Program Participant (TACC, Minneapolis, July 2026); UC Berkeley Fung Fellow (Spring 2026)
- Presented at SCCUR (San Bernardino, CA, November 2024) and STARS Celebration Conference (San Diego, CA, September 2024)